Geo_DECT

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Installation

Dragonfly plugin for stoichiometric DECT processing (Z_{eff} / ρ_e / combined visualization, 3D rendering, statistics export).

1. Requirements

- Dragonfly 3D 2025.1 (or compatible) already installed.
- Administrator access to the machine is **not** required; the plugin installs into your user profile.

2. Files to copy

Copy the **entire** Geo_DECT_7e90f2c77eb011f18afab07b2518f399 folder, containing exactly these files:

```
Geo_DECT_7e90f2c77eb011f18afab07b2518f399/
├── Geo_DECT_7e90f2c77eb011f18afab07b2518f399.py
├── Geo_DECT_7e90f2c77eb011f18afab07b2518f399.qrc
├── __init__.py
├── geodect_3d.py
├── geodect_core.py
├── geodect_statistics.py
├── geodect_visualization.py
├── mainformgeo_dect.py
└── ui_mainformgeo_dect.py
```

Do **not** rename the folder or the .py/.qrc files matching the folder name, Dragonfly identifies the plugin by that exact name.

Do **not** use Dragonfly's "Python Plugin Generator" to create a new plugin folder for this: it assigns a new random ID every time, which does not match the ID already used inside mainformgeo_dect.py (WorkingContext.registerOrsWidget(...)), and the plugin silently fails to appear with no error message. Always copy this exact, already-working folder as-is.

3. Copy the plugin into Dragonfly 3D

1. Close Dragonfly 3D if it is running.
2. Open File Explorer, click the address bar, and paste: (or press Windows key + R and paste it in the Run window; the Windows search bar opens a web search instead)
3. `%localappdata%\Comet\Dragonfly2025.1\pythonUserExtensions\Plugins\`

   Press *Enter*. This works on any machine, Windows fills in the correct username automatically.

   Dragonfly2025.1 must match the installed version: if %localappdata%\Comet contains a different folder (e.g., *Dragonfly2024.1*), use that one. For an installation shared by all accounts on the

machine, use %ProgramData%\Comet\Dragonfly2025.1\pythonAllUsersExtensions\Plugins\ instead.

4. Paste the whole Geo_DECT_7e90f2c77eb011f18afab07b2518f399 folder there, so the path looks like:

```
...\Plugins\Geo_DECT_7e90f2c77eb011f18afab07b2518f399\mainformgeo_dect.py
```

The .py and .qrc files must stay inside the Geo_DECT_7e90f2c77eb011f18afab07b2518f399 folder (Figure 1): not loose in Plugins, and not in a second folder with the same name (this often happens when extracting a .zip file).

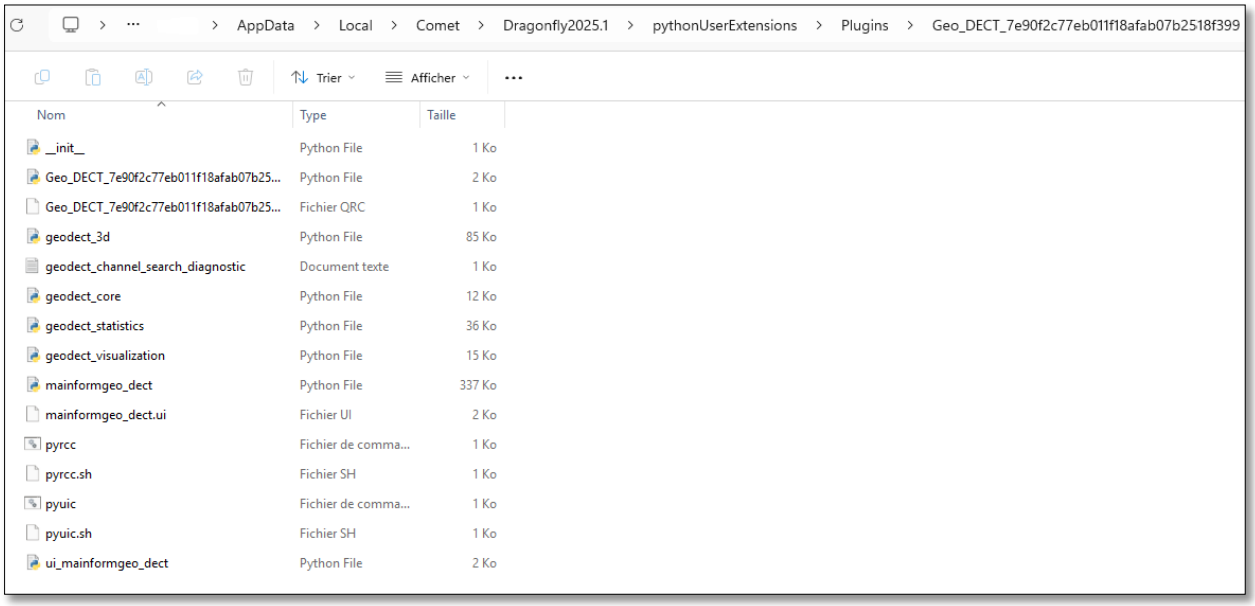


Figure 1: Correct location of the plugin: the Geo_DECT_7e90f2c77eb011f18afab07b2518f399 folder inside Plugins, with the .py and .qrc files directly inside it.

4. Start Dragonfly 3D

1. Launch Dragonfly 3D.
2. Go to **Utilities → Open Plugins → Geo_DECT**. If it does not appear there, double-check step 3: the folder name and its location must match exactly.

5. Troubleshooting

- **Plugin does not appear under** Utilities **→ Open Plugins**: wrong folder name/location, files copied loose into Plugins instead of inside their folder (Figure 1), the folder was generated with Dragonfly's own Plugin Generator instead of being copied as-is (see the warning in step 2), or Dragonfly was not restarted after copying the files.
- **"Export statistics" / "Export crossplot" fails with an error mentioning matplotlib**: this package is missing from Dragonfly's own Python environment. Open a command prompt and run (adjust the path if Dragonfly is installed elsewhere): "C:\Program

Files\Dragonfly\Python_env\python.exe" -m pip install matplotlib". Every other part of the plugin works normally without this step.

- **Any other error on startup**: check Dragonfly's own log/console for a Python traceback and report it, rather than guessing.
- **"Channel creation failed"** popup during Build: it appears if creating the Z_{eff} or ρ_e volume fails. It saves a diagnostic report (array shape, NaN/Inf counts, min/max) next to the plugin files (*geodect_channel_diagnostic.txt*) and shows it directly in the popup. This most often means the calibration is being extrapolated well outside the range it was built for (see the next point).
- On very large datasets (several billion voxels), the same popup can mean that the RAM ran out. Crop the bounding box, or bin both stacks 2×2×2 with Average (never Sum) in ImageJ/Fiji and double the value in Voxel size override. Keeping the dataset below about 2 billion voxels is recommended.
- **Some voxels stay transparent within the sample, even though the mask looks correct**: Z_{eff} outside 0–50 or ρ_e outside 0–1×10^{24} $e^{-}\cdot cm^{-3}$ is automatically excluded rather than shown, since a calibration curve evaluated far outside the range it was fitted on can otherwise produce meaningless values. If this happens for many voxels, or across a large connected region rather than scattered individual points, it is worth checking whether the calibration in use actually covers the physical range this sample produces (e.g., by comparing the μ_{low}/μ_{high} ratio of the sample against the ratio range of the calibration's own reference materials). Voxels whose μ_{LOW}/μ_{HIGH} ratio falls outside the plausible range set in the calibration panel (Usage Guide, section 3) are excluded as well, before the calibration is applied.

Usage Guide

Step-by-step workflow, from loading the two CT datasets to the final 3D volumes.

1. Load the datasets

At the top of the panel, browse to the LOW-energy and HIGH-energy folders. Both datasets are loaded once selected (Figure 2).

2. Check datasets and calibration

Click **Check datasets and calibration** (Figure 2). This step is mandatory: *Geo_DECT* determines automatically whether it can calculate the linear attenuation coefficients μ on its own, or whether it needs manual input from you.

Right below it, *Reset current analysis* clears everything entered so far in this panel (calibration, mask, test results) so you can start over cleanly on a new dataset, without closing and reopening the plugin.

- If manual input is needed, the min/max fields for the two stacks become editable. Otherwise they stay locked.
- When editable, enter the **physical** min/max values of the LOW and HIGH energy stacks (**not** the display/normalized values!).

Right below the *Check* button, a Voxel size override (μm) field lets you enter (or correct) the physical size of one voxel, in micrometres, either one value (isotropic voxel, applied to X, Y and Z alike, typical for micro-CT) or three values X, Y, Z (anisotropic voxel, typical for medical DICOM, where the slice spacing differs from the pixel size). Every time you click *Check* datasets and calibration, a popup shows the voxel size detected from the file's own metadata, pre-filled into an editable box: check it against your scan/acquisition software and correct it right there if it looks wrong. If a file carries no physical metadata at all (this can happen with some acquisition software), the popup says so explicitly instead of proposing a number: in that case the value must always be entered by hand, since there is nothing in the file to detect it from. Once entered, the override value always takes priority over auto-detection, for every file format.

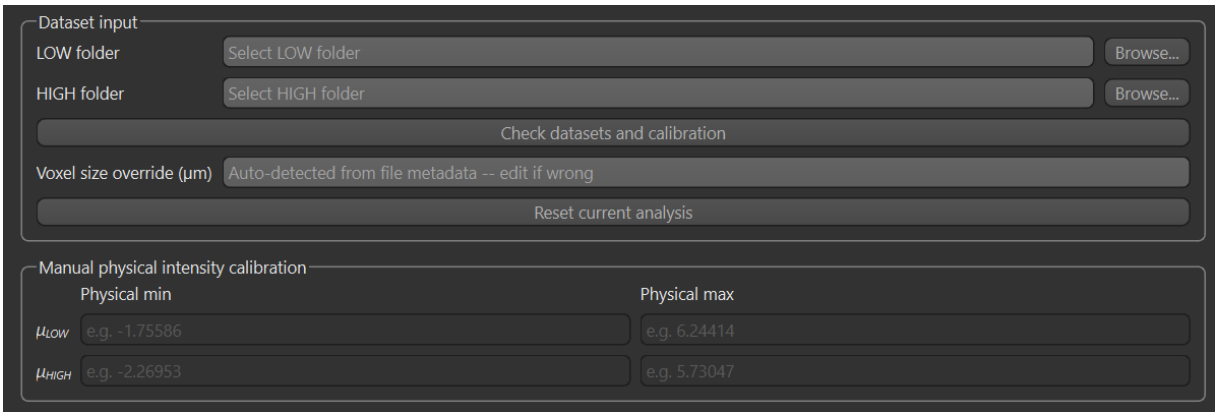


Figure 2: The Dataset input panel after loading the LOW and HIGH folders, with the Check datasets and calibration button and the Voxel size override field.

3. Calibration

1. Enter the **polynomial degree** used for the calibration (fields become editable up to the chosen degree, maximum degree 5).
2. Enter the corresponding **coefficients**.
3. Choose which μ to use for the electron density (ρ_e) calculation: **low energy**, **high energy**, or the **average** of both.
4. Leave **Convert ρ_e / $\rho_{e,water}$ to absolute $e^{-}\cdot cm^{-3}$** checked if the calibration returns ρ_e relative to water (values of a few units): Geo_DECT detects these values and multiplies them by $\rho_{e,water} = 3.343\times10^{23}$ $e^{-}\cdot cm^{-3}$. Calibrations that already return absolute values are left unchanged.
5. **μ_{LOW}/μ_{HIGH} plausible range** (Minimum, Maximum; prefilled 0.9 and 1.4): voxels whose ratio falls outside it are excluded before the polynomial is applied, so that partial-volume voxels or noise do not produce misleading Z_{eff} and ρ_e values. Set it to the ratio range of your calibration materials, with a margin; leave both fields empty to disable this check.
6. **Noise reduction before the μ_{LOW}/μ_{HIGH} ratio**: None (raw voxels), Gaussian, mask-aware (recommended; Sigma in pixels, default 1.5, 1-2 is usually enough) or Median (Window in pixels, odd, at least 3). The same filter is applied to μ_{LOW} and μ_{HIGH}, slice by slice, and only the voxels retained by the mask are used, so the background does not affect the sample edge. The filter applies to the test slice, the 3D volumes and the statistics.
7. Optionally **save** the calibration coefficients under a custom name, to reuse them later without re-entering everything (Profile name, **Save current coefficients**); saved profiles are recalled with **Load selected profile** and removed with **Delete selected profile**. The last profile used is restored automatically when Geo_DECT starts.
8. Click **Validate coefficients and show equations** and the final equations used for the calculation are displayed (see Figure 3).

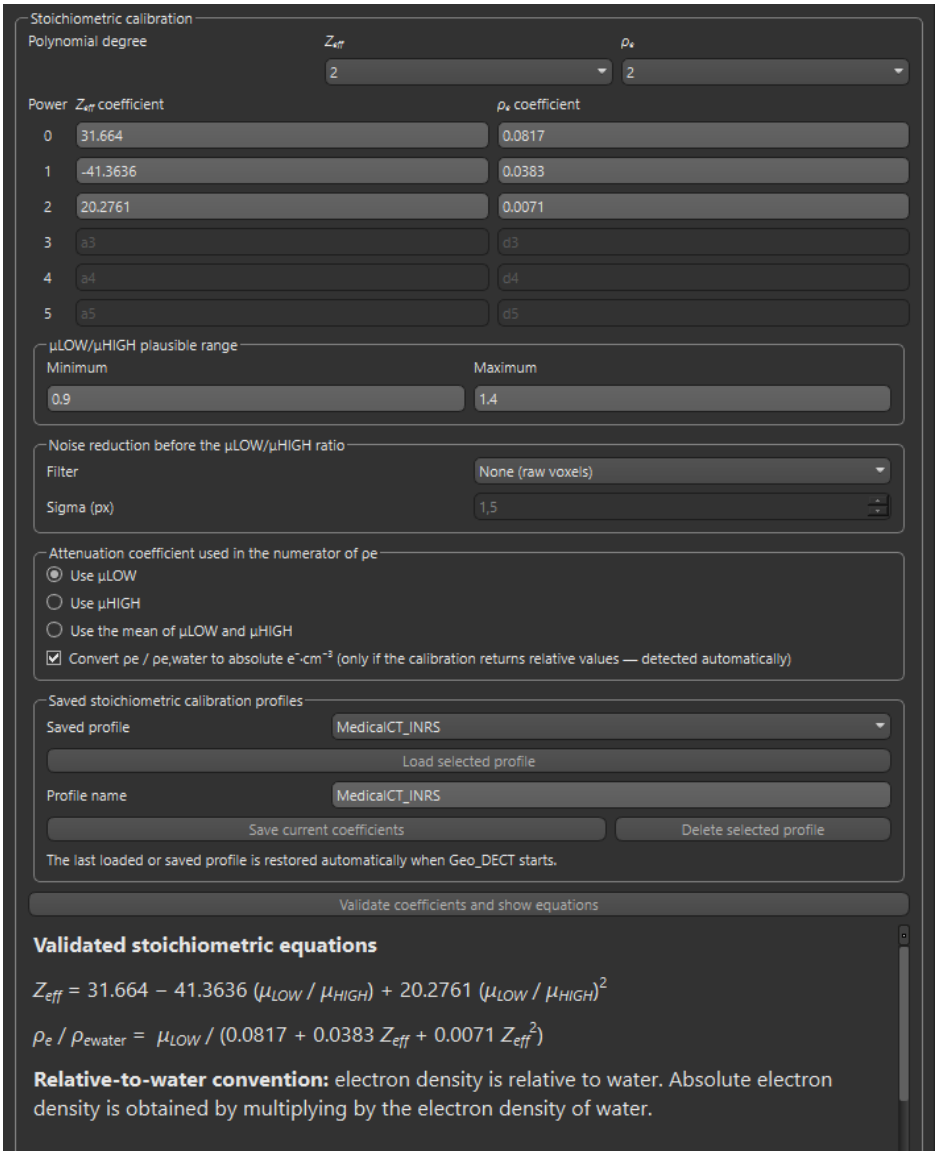


Figure 3: The calibration panel: polynomial degree and coefficients, μ_{LOW}/μ_{HIGH} plausible range, noise reduction, attenuation coefficient used for ρ_e, saved profiles and validated equations.

4. Define the region to analyze (mask)

Geo_DECT processes only the voxels retained by the mask. The mask settings depend on the sample, the scanner and the acquisition, so they must always be adjusted to each dataset.

Exclude background keeps only the voxels whose HIGH-energy value is above the threshold (HIGH HU threshold for DICOM, HIGH gray-level threshold for TIFF). Choose a threshold that separates the sample from air and, if present, from the core liner or sample holder.

Mask erosion (px) shrinks the mask inward, slice by slice, and removes the thin partial-volume rim at the sample edge and around pores (0 = no erosion). Increase it step by step and check the preview each time: too much erosion also removes small pores and thin features.

Click **Preview retained voxels**: the retained voxels are highlighted in purple/lilac (Figure 4). Click the preview to enlarge it. Repeat until the purple area covers the sample and nothing else.

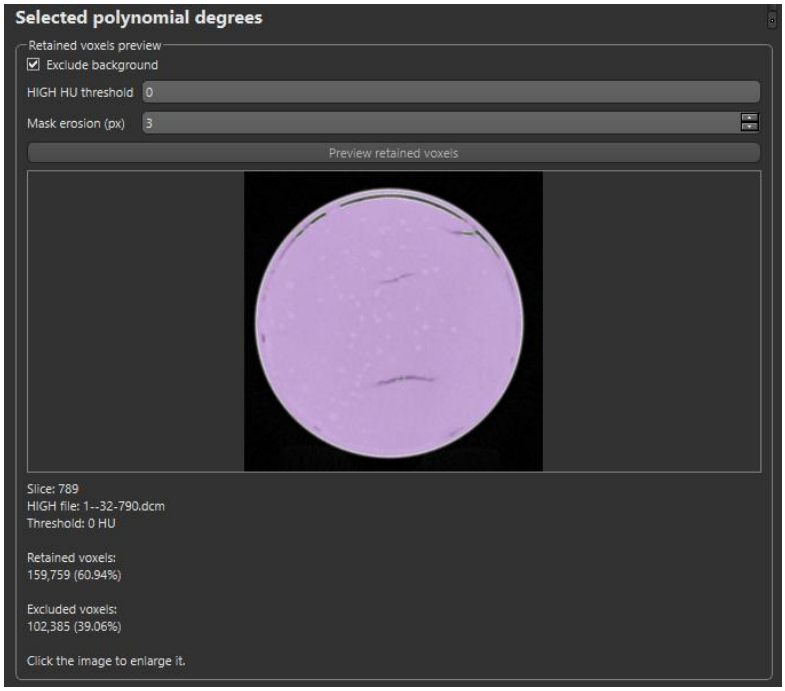


Figure 4: The mask panel: background threshold, mask erosion and preview with the retained voxels in purple/lilac.

5. Preliminary test

1. Choose the test slice: **middle**, **first**, or a **custom** one.
2. Click **Test calculation on selected image pair**: the first calculation runs on that single slice.
3. The detected Z_{eff} and ρ_e min/max values are filled in automatically (**Detect automatic ranges from calculated slice** recomputes them); the first reconstruction preview uses these (Figure 5).
4. All these fields stay editable, adjust them if needed.
5. **Create visualization previews** shows the expected result (Figure 6).
6. **Save test-slice maps as 32-bit TIFF...** saves Z_{eff}, ρ_e, the raw μ_{LOW}/μ_{HIGH} ratio, μ_{LOW}, μ_{HIGH} and the mask of the test slice, for independent checks in ImageJ/Fiji.

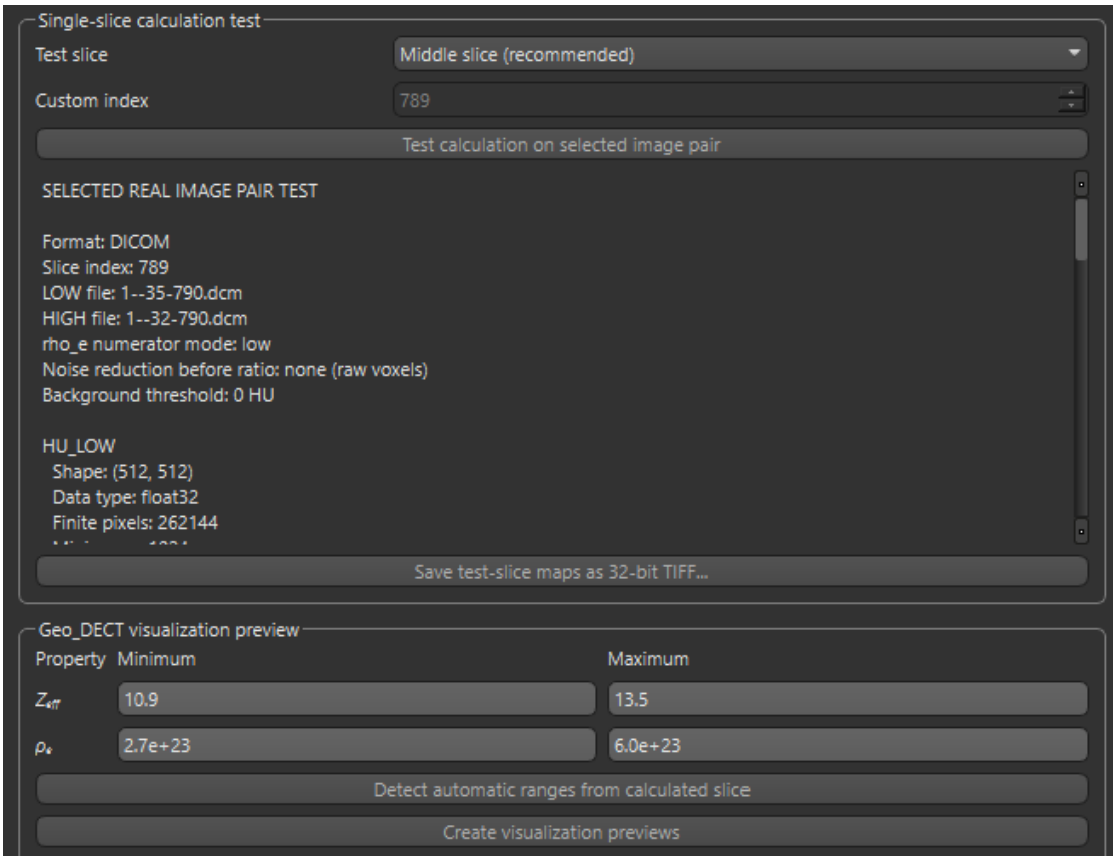


Figure 5: the test panel after running the calculation, showing the detected Z_{eff} and ρ_e min/max.

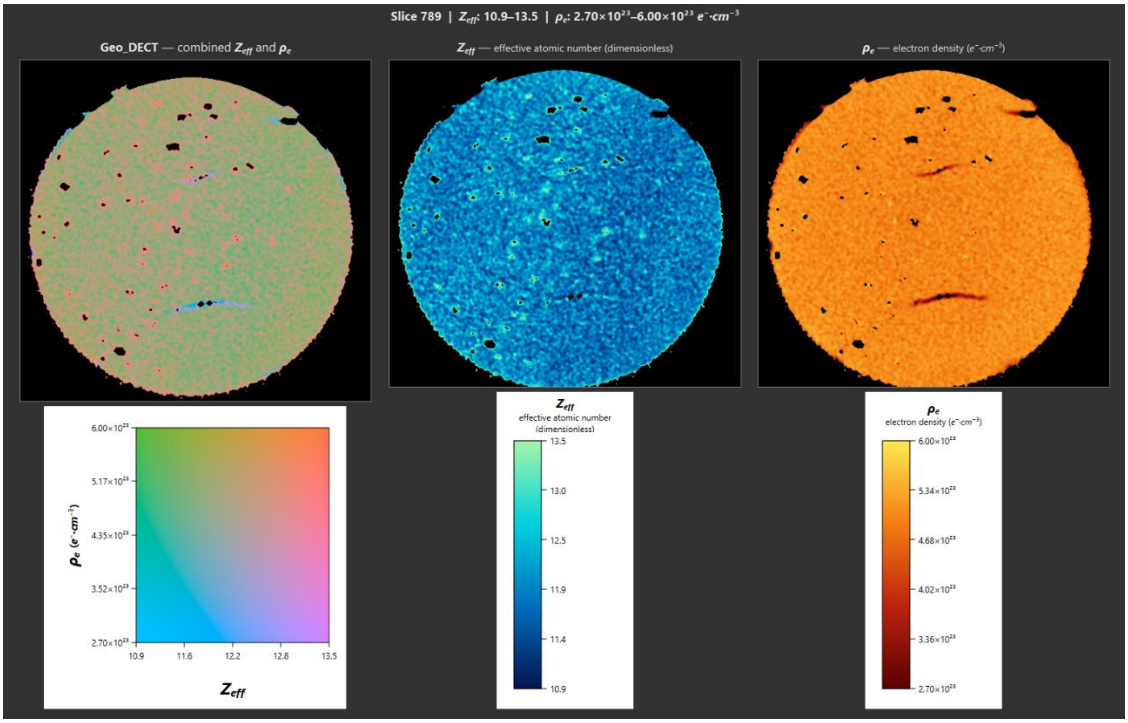


Figure 6: The preview of the expected result.

6. Combined colour resolution

Choose the number of colours for the combined (Z_{eff} and ρ_e) volume: **~4,900**, **~10,000**, or **~14,900**. A higher number gives a finer colour gradient but a heavier volume.

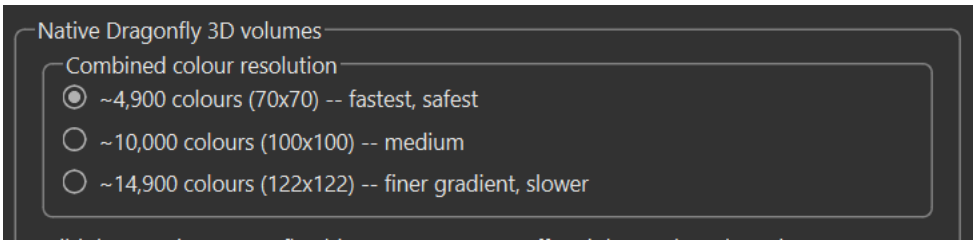


Figure 7: The resolution selector.

7. Build the 3D volumes

Click **Build and publish 3D volumes**. A message reports the disk space required before proceeding. Once confirmed, the three volumes (Z_{eff}, ρ_e, combined) are created and behave as normal native Dragonfly 3D objects from that point on; every standard Dragonfly 3D tool works on them directly (a message confirming the three native channels are available appears).

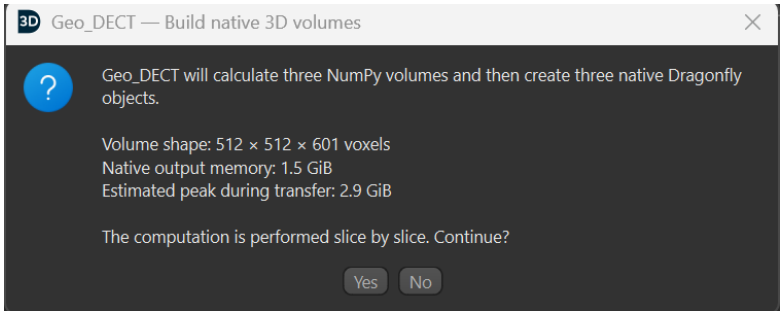


Figure 8: The space-required confirmation message, once the three resulting volumes in Dragonfly's object are built.

If a volume appears **grey** right after building, click **Force 3D coloring**: it appears right below the "*Build and publish 3D volumes*" button and its progress bar. The 3D renderer sometimes needs a manual nudge right after the volume is created.

After reopening a saved Dragonfly project, Geo_DECT reconnects its volumes automatically when the plugin opens. If the colours or the 3D volume do not come back, click **Reconnect Geo_DECT volumes (after reopening a project)**.

8. Exploring Z_{eff} and ρ_e after building

A dedicated **"*Geo_DECT* display range"** panel appears on the right. For Z_{eff} and ρ_e independently:

1. Pick a sub-range with the slider (or type it directly). **Full stack range** restores the complete range.
2. Click **Apply**; the volume is instantly recoloured, both in 2D and in 3D, restricted to that range (everything outside it becomes transparent).
3. The colour scale shown next to the sliders always reflects the **true, correctly rescaled physical values** for that property (including the $\times 10^{23}$ unit for ρ_e); this is the reference to read values from, since Dragonfly 3D's own native scale bar/legend on these volumes is deliberately hidden by the plugin (it would otherwise show internal colour codes, not the true physical values), not seeing one there is expected, not a sign anything is wrong.
4. Click on that colour scale to enlarge it and save it as an image.

This can be repeated as many times as needed to explore different sub-ranges of the sample.

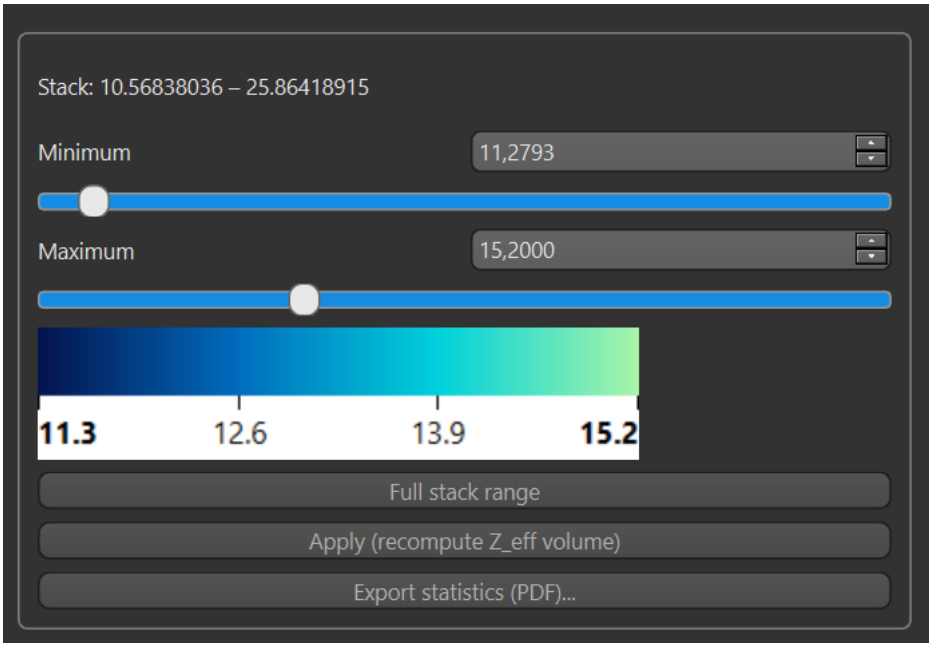


*Figure 9: The "Geo_DECT display range" panel for Z_{eff} (ρ_e is in shades ranging from red to yellow), with a sub-range selected and **Apply** already clicked, showing the recoloured volume next to it.*

9. Exploring the combined volume

The combined (Z_{eff} and ρ_e) volume works the same way, through the colorimetric scatterplot:

1. Select the Z_{eff} and ρ_e range you want, either by dragging the dashed rectangle over the region of interest on the scatterplot (move it from the middle, resize it from an edge/corner), or by typing the exact minimum/maximum values directly into the four "Manual range entry" fields below it (one pair for Z_{eff}, one for ρ_e, each with its own slider); both are kept in sync, so the rectangle moves as you type or drag the sliders, and the fields update as you drag the rectangle.
2. Click **Apply** (recompute combined volume); the combined volume is recoloured accordingly.
3. The square next to the scatterplot always previews the colours that particular sub-range produces once redistributed, with the exact Z_{eff} and ρ_e boundaries printed underneath it.

4. **Enlarge / Save...** produces one image with both the scatterplot and the redistribution preview together, ready to use in a figure or a presentation (all images can be saved separately).

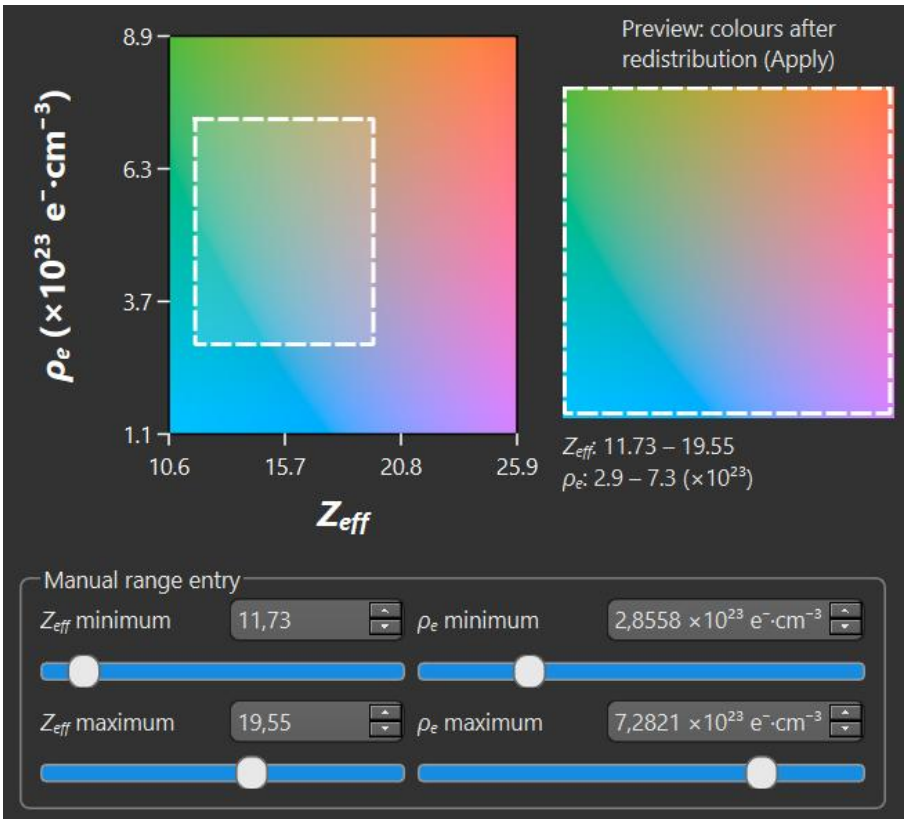


Figure 10: The scatterplot with a selection rectangle drawn, the manual range entry fields (with sliders) below it, and the redistribution preview next to it, showing the applied Z_{eff} and ρ_e boundaries.

10. Using Probe

Dragonfly's own **Probe** tool is very useful here: click anywhere on the Z_{eff}, ρ_e, or combined volume, and a dot is added automatically to the combined scatterplot at that point's exact Z_{eff} / ρ_e position. Every probed point is also listed in the "Probed points" table below the scatterplot buttons (Z_{eff} and ρ_e values, in full physical units), which can be exported as a CSV file with Export CSV..., or cleared with Clear points.

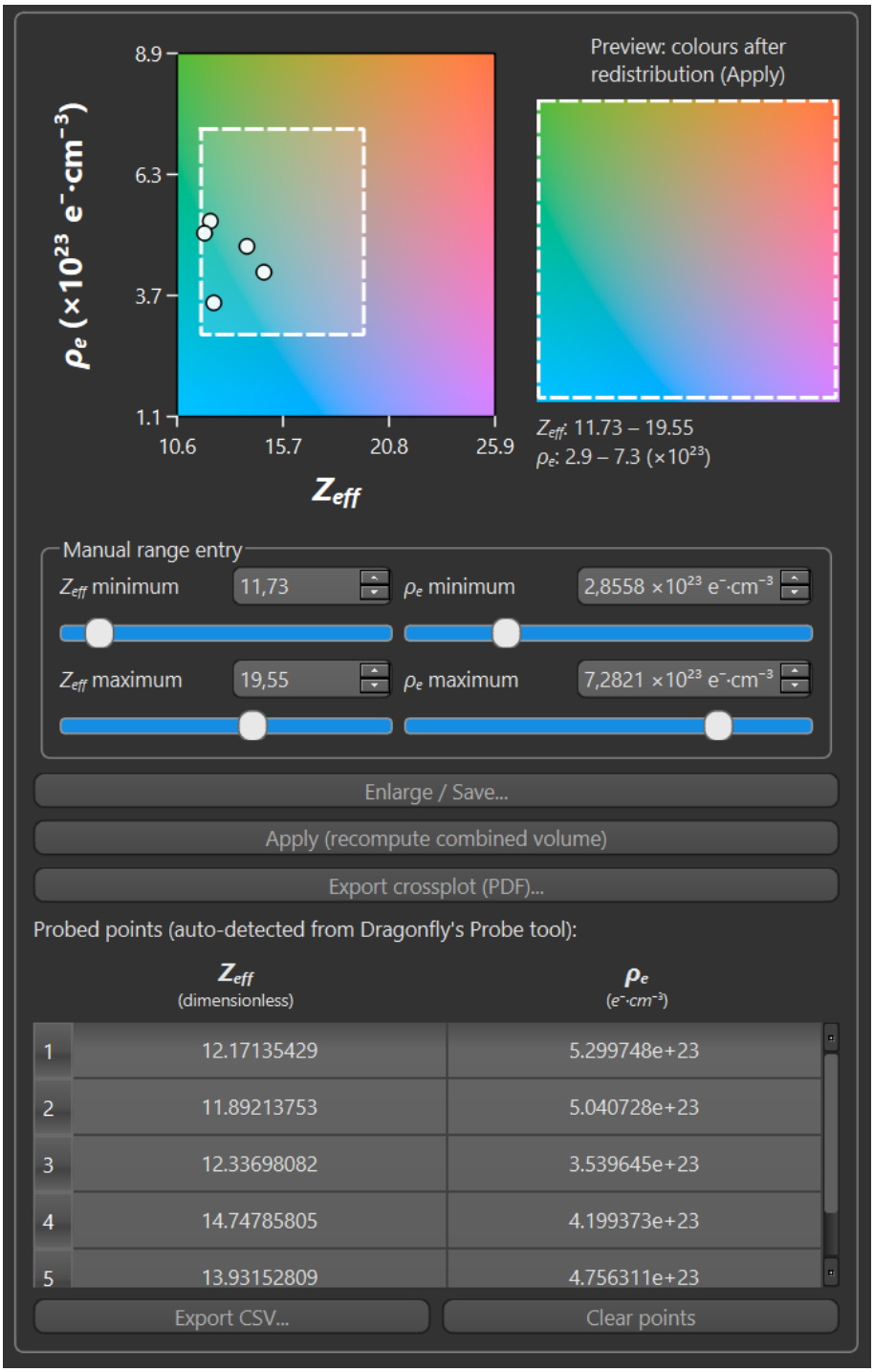


Figure 11: A Probe click on the volume next to the resulting dot on the scatterplot, and the points table below it.

11. Statistical analysis

Both individual properties and the combined volume can be exported as ready-to-use figures, always based on the **range most recently confirmed with Apply**:

- ***Z_{eff} / ρ_e*: Export statistics (PDF)...**

A dialog opens first: **Number of populations**, Automatic (BIC, 1-3) or 1, 2, 3; **Exclude interface (partial-volume) voxels** (checked by default), which removes, slice by slice, the given percentage of voxels with the strongest local gradient (default 30%), i.e. voxels on phase boundaries (Figure 12).

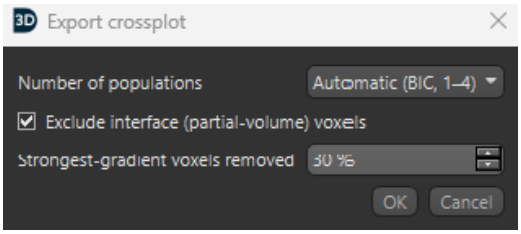


Figure 12: The options dialog opened by Export statistics (PDF)... and Export crossplot (PDF)....

Every valid voxel of the sample is first split into three groups: inside, below, and above the selected range.

All the numbers and shapes below use only the **inside group**:

- mean, median, standard deviation, min, max
- mode → 200-bin histogram, centre of the tallest bin
- percentiles P1/P5/P25/P75/P95/P99
- the grey histogram bars (as a probability density, not raw counts)
- the Gaussian mixture: one shaded curve per population (mean ± standard deviation and volume fraction in the legend) and their sum (black curve); it is fitted on a random subsample of up to 100,000 voxels, and the populations are also listed in the summary box
- the boxplot (box = P25–P75, red line = median, whiskers = 1.5×IQR; individual outlier markers are not drawn, to keep large volumes fast to export; the boxplot is drawn from a random subsample of up to 200,000 values, while the percentiles use all inside voxels)

The voxel counts for inside/below/above the range (and their percentages) are also reported.

The PDF opens automatically once generated (Figure 13).

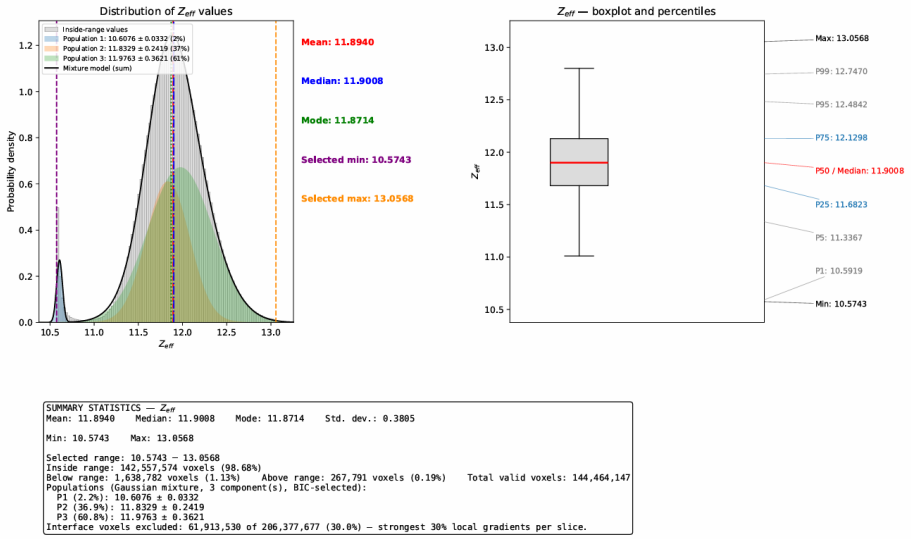


Figure 13: An example of the exported statistics PDF for Z_{eff}.

- **Combined**: **Export crossplot (PDF)...**

  Only voxels whose Z_{eff} AND ρ_e both fall inside their respective selected ranges are used. From those: interface voxels are excluded as for Z_{eff} / ρ_e; here a voxel is removed if its gradient is among the strongest in Z_{eff} or in ρ_e.

  - A 2D density plot (voxel count per cell) of Z_{eff} *vs* ρ_e. The colour of each cell gives the number of voxels (logarithmic scale), and the Z_{eff} and ρ_e histograms are shown along the axes.
  - Populations: *Geo_DECT* describes the cloud of points as the sum of one to three Gaussian populations (two-dimensional Gaussian mixture model). The number of populations is chosen automatically with the Bayesian Information Criterion (BIC), or set in the dialog (Figure 12). The model is fitted on a random subsample of up to 100,000 voxels, which keeps the export fast on large volumes. Each population is drawn as two ellipses, 1σ (solid) and 2σ (dashed), and its mean is marked by ×. Populations are numbered from the lowest to the highest mean Z_{eff} (P1 = lowest). Each population also appears as a coloured curve on the Z_{eff} histogram (top) and on the ρ_e histogram (right), in the same colour as its ellipses.
  - The summary below the plot lists, for each population, percentage of voxels and the mean ± standard deviation of Z_{eff} and ρ_e.

  The PDF opens automatically once generated (Figure 14).

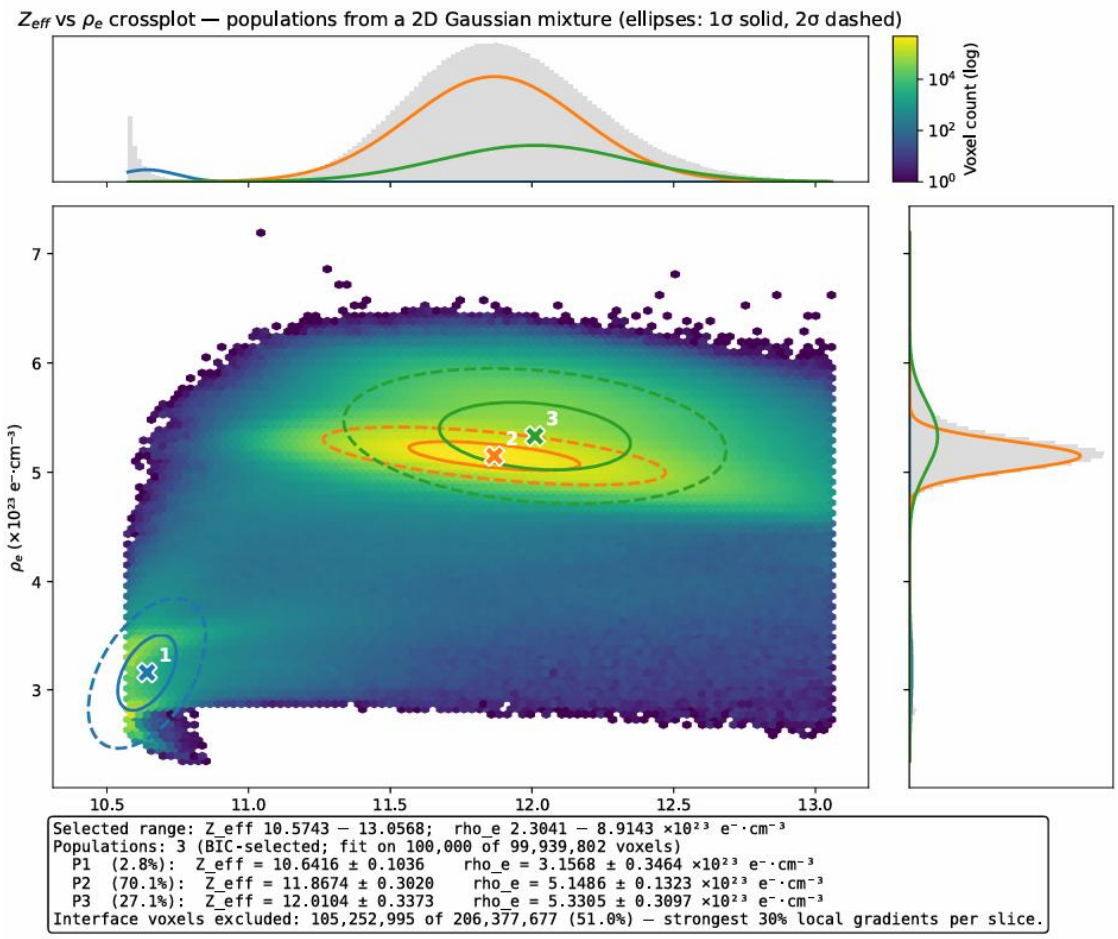


Figure 14: An example of the exported combined crossplot PDF.

Additional Information

Summary of outputs, example data, background references, previous presentations, license and support.

1. Outputs at a glance

All files and volumes that Geo_DECT produces, with the step of the Usage Guide where each one is created:

Output	How to obtain it	Format
Z_{eff}, ρ_e and combined volumes	**Build and publish 3D volumes** (Usage Guide, section 7)	Native Dragonfly 3D channels
Test-slice maps (Z_{eff}, ρ_e, μ_{LOW}/μ_{HIGH} ratio, μ_{LOW}, μ_{HIGH}, mask)	**Save test-slice maps as 32-bit TIFF…** (section 5)	32-bit TIFF
Colour scale of Z_{eff} or ρ_e	Click on the colour scale, then save (section 8)	Image
Scatterplot with redistribution preview	**Enlarge / Save…** (section 9)	Image
Probed points	**Export CSV…** (section 10)	CSV
Z_{eff} / ρ_e statistics	**Export statistics (PDF)…** (section 11)	PDF
Combined crossplot	**Export crossplot (PDF)…** (section 11)	PDF
Channel diagnostic report	Written automatically if channel creation fails (Installation, section 5)	Text file (geodect_channel_diagnostic.txt)

2. Example data

The dual-energy CT datasets of the two case studies presented in the *Geo_DECT* article (an oolitic hematite ironstone scanned by micro-CT and a varved lacustrine sediment core scanned by medical CT) are available on Zenodo, together with the calibration coefficients and the *Geo_DECT* parameters used for each sample. They can be used to test the plugin and to reproduce the results of the article:

https://doi.org/10.5281/zenodo.24105517

3. Background references

The theoretical basis of the stoichiometric DECT calibration, from which the coefficients entered in *Geo_DECT* are derived, is described in Martini et al. (2021). Martini et al. (2025) compares the calibration and its results on three different scanners. Both articles give further details on calibration and instrument set-up, and can be cited for these aspects:

- Martini, M., Francus, P., Di Schiavi Trotta, L., Després, P., 2021. Identification of common minerals using stoichiometric calibration method for dual-energy CT. *Geochemistry, Geophysics, Geosystems* 22(11), e2021GC009885. https://doi.org/10.1029/2021GC009885
- Martini, M., Francus, P., Schröer, L., Buyse, F., Kibleur, P., Cnudde, V., Di Schiavi Trotta, L., Després, P., 2025. Characterisation of geological samples with dual-energy XCT: a comparison of

three different scanners. *Geostandards and Geoanalytical Research* 49(3), 645–664. https://doi.org/10.1111/ggr.12608

4. Previous presentations

Geo_DECT has been presented at the following meetings; the first two showed an early Python prototype of the plugin:

- Martini, M., Francus, P., Letellier, P., 2025. *Geo_DECT*: a new tool to visualize dual-energy CT data and study lacustrine sediments. *IAL–IPA 2025 Joint Meeting*.
- Martini, M., Colombié, C., Rinder, N., Slater, G.F., Hartz, J., Sahakyan, L., Joannin, S., 2026. Unlocking microbialite structure with dual-energy CT. *International Conference on Tomography of Materials and Structures (ICTMS 2026)*, Manchester, UK.
- Francus, P., Martini, M., 2026. Micro-computed X-ray tomography. *PaleoImaging Workshop 2026*, University of Gdańsk, Poland.

5. License

Geo_DECT is distributed under the MIT License (see the LICENSE file in the repository). Dragonfly 3D is developed by Comet Technologies and is not part of this repository: it requires its own license, which the Geo_DECT license does not cover.

6. Support and bug reports

Questions, bug reports and suggestions are welcome on the Issues page of the GitHub repository:

https://github.com/martinimargheritamartini-dotcom/Geo_DECT/issues

A report is much easier to solve when it includes:

- the Dragonfly 3D version and operating system;
- the dataset size (number of voxels) and file format (TIFF or DICOM);
- the Python traceback from the Dragonfly log/console, if any;
- the geodect_channel_diagnostic.txt file, if the “Channel creation failed” popup appeared.